```python
# Install OpenCV (if needed)
!pip install opencv-python

# Import required libraries
import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()

# Get uploaded file name
image_path = list(uploaded.keys())[0]

# Read the image
image = cv2.imread(image_path)

# Check if image is loaded
if image is None:
    print("Error: Unable to load image.")
else:
    # Convert BGR to RGB for display
    image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

    # Convert to grayscale
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

    # Apply Gaussian Blur to reduce noise
    blurred = cv2.GaussianBlur(gray, (5, 5), 0)

    # Apply Sobel Edge Detection along Y-axis
    sobel_y = cv2.Sobel(
        blurred,
        cv2.CV_64F,   # Output image depth
        0,            # dx = 0 (No X-direction)
        1,            # dy = 1 (Y-direction)
        ksize=3       # Kernel size
    )
```

```
# Convert to absolute values
sobel_y = cv2.convertScaleAbs(sobel_y)

# Display the images
plt.figure(figsize=(12,6))

plt.subplot(1,2,1)
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(sobel_y, cmap='gray')
plt.title("Sobel Edge Detection (Y-axis)")
plt.axis("off")

plt.show()

# Save the output image
output_path = "sobel_y_output.png"
cv2.imwrite(output_path, sobel_y)

print("Sobel Y edge detection completed successfully!")
```

```
Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-packages (5.0.0.93)
Requirement already satisfied: numpy>=2 in /usr/local/lib/python3.12/dist-packages (from opencv-python) (2.0.2)
```

Choose Files Screenshot… 231222.png

Screenshot 2026-08-16 231222.png(image/png) - 880749 bytes, last modified: 16/8/2026 - 100% done

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Saving Screenshot 2026-08-16 231222.png to Screenshot 2026-08-16 231222.png
```

Original Image

Sobel Edge Detection (Y-axis)

```
Sobel Y edge detection completed successfully!
```